

Specification of Displayer Module

Customer Project Name	
Module Name	AML055F4001
Version No.	V00 2024-11-6

Customer Approval:

- ☐ Approve Specification Only
- ☐ Approve Specification and Sample

Approved By
Data

Designed By	Checked By	Approved By

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1. Introduction

This displayer module is designed by Shenzhen Amelin Electronic Technology Co., Ltd.. in Shenzhen China. All materials and processes of this displayer module are Lead Free.

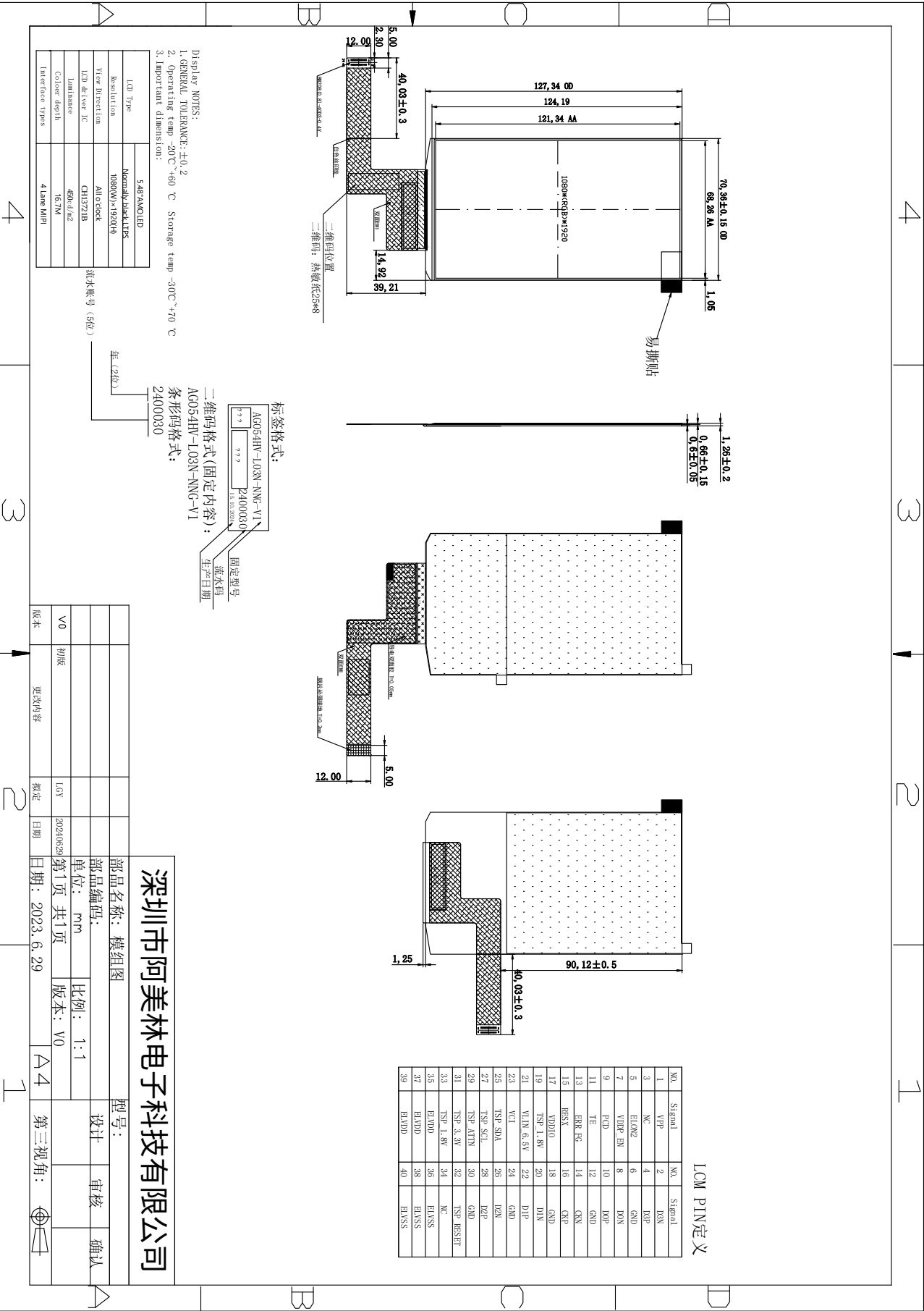
Displayer, displayer module, LCM or module in this specification has the same meaning.

2. General Specification

2.1 Displayer Module General Specification

Item	Content	Unit
Displayer Module Size (inch)	5.48	inch
Active Area (W×H)	68.256×121.344	mm
Module Size (W×H×T)	70.36×127.34×1.26	mm
Displayer Type	AMOLED	/
Recommended Viewing Virection	ALL	/
Number of Dots	1080×(RGB)×1920	Dots
Pixel Per Inch (PPI)	402	/
Driver IC	CH13721C	/
Interface Type	MIPI	/
Approx. Weight	TBD	g
NTSC	99%(Type)	/

3.Mechanical Drawing



4. Displayer Module Absolute Maximum Ratings

Item	Symbol	Min.	Max.	Unit
Supply Voltage for Logic	VDDI	-0.3	3.3	V
DC/DC Converter Circuit Output	VCC	2.8	3.3	V
Driver Supply Voltage	VGH/VGL	-0.3	+30.0	V
Input Voltage for Logic	Vt	-0.3	VDDIO+0.3	V
Operating Temperature	Top	-20	+70	℃
Storage Temperature	Tst	-30	+80	℃

5. Electrical Characteristics

5.1 Displayer Module Electrical Characteristics

Item	Symbol	Min.	Typ.	Max.	Unit
Supply Voltage for Logic	VDDIO	1.65	1.8	3.3	V
Supply Voltage for Analogic	VCI	2.8	3.2	3.3	V
Input Voltage	V _{IL}	V _{SS}	-	0.3Vddi	V
	V _{IH}	0.7VDDI	-	VDDI	V
Input Leakage Current	I _{LKG}	-0.1	-	0.1	μA

6. Displayer Module Optical Characteristics

Item		Symbol	Condition	Specifications			Unit	Remark
				Min.	Typ.	Max.		
Brightness		Lv	Viewing normal angle $\Theta=0^\circ$	420	450	--	Cd/m2	Note 2
Contrast Ratio		CR		60000	--	--	/	Note 1
Response Time		Tr+Tf		--	--	2	ms	Note 4
Chromaticity CIE(x,y)	Red	x		0.682±0.03			/	Note 6
		y		0.315±0.03			/	
	Green	x		0.24±0.03			/	
		y		0.716±0.03			/	
	Blue	x		0.138±0.03			/	
		y		0.048±0.03			/	
	White	x		0.275±0.03			/	
		y		0.275±0.03			/	
Viewing Angle	Hor.	θ_{x+}	Center CR>=10	--	85	--	Deg.	Note 5
		θ_{x-}		--	85	--		
	Ver.	θ_{y+}		--	85	--		
		θ_{y-}		--	85	--		
Luminance Uniformity		White		75	--	--	%	Note 3

Note1. Contrast Ratio (CR) is defined mathematically by the following formula. For more information see FIG.1

$$\text{Contrast Ratio} = \frac{\text{Average Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)}}{\text{Average Surface Luminance with all black pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)}}$$

Note2. Surface luminance is defined mathematically by the following formula. For more information see FIG.1

$$\text{Brightness} = \text{Average Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)}$$

Note3. The luminance uniformity is defined mathematically by the following formula. For more information see FIG.1

$$\text{Luminance Uniformity} = \frac{\text{Minimum Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)}}{\text{Maximum Surface Luminance with all white pixels (P1,P2,P3,P4,P5,P6,P7,P8,P9)}}$$

Note4. Response time is the time required for the displayer to transition from White to black(Rise Time, Tr) and from black to white(Decay Time, Tf). For more information see FIG.2

Note5. Viewing angle means the best contrast ratio of displayer for user. For more information see FIG.3

Note6. Chromaticity CIE(x, y) is determined by the CIE(x, y) value of the point5. For more information see FIG.1

Note7: NTSC ratio is defined mathematically by the following formula. For more information see FIG.4

NTSC = $\frac{\text{Area of RGB triangle}}{\text{Area of NTSC triangle}}$

Ratio

Note8. For viewing angle and response time testing, the testing data is base on instruments of Autronic-Melchers's ConoScope series. For contrast ratio, Surface Luminance, Luminance uniformity and CIE(x,y), the testing data is base on BM-7 photo detector.

Note9. For TFT transmissive module, change the viewing angles could see gray scale reverse.

FIG.1. Measuring method for Contrast ratio,surface luminance, Luminance uniformity,CIE(x,y) chromaticity

A: 6 mm

B: 6 mm

H,V: Active Area

Light spot size $\Phi=5\text{mm}$, 500mm distance from the displayer surface.

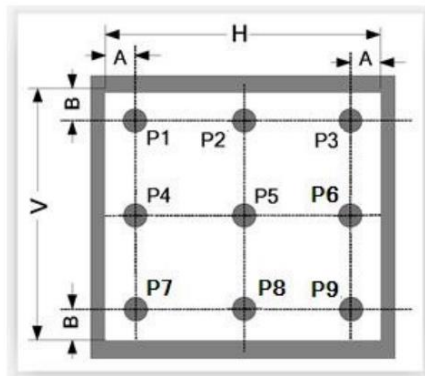


FIG.1. 9-point position on displayer 测量点位置示意

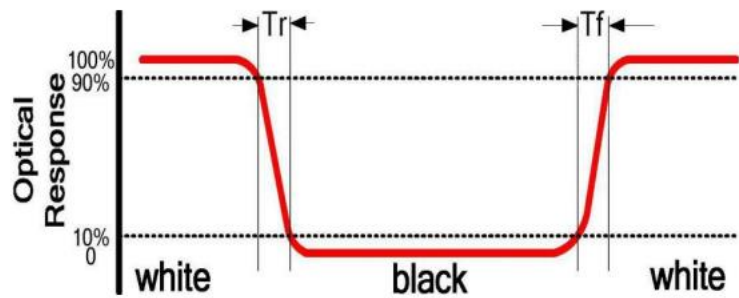


FIG.2. The definition of response time 响应时间定义

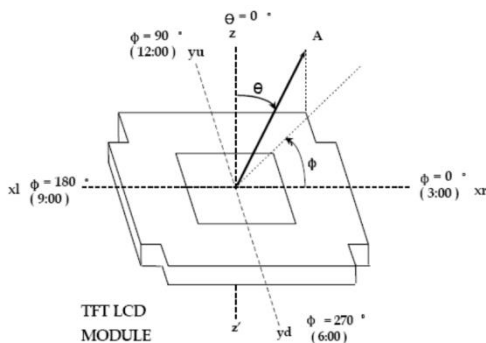


FIG.3. The definition of viewing angle 视角定义

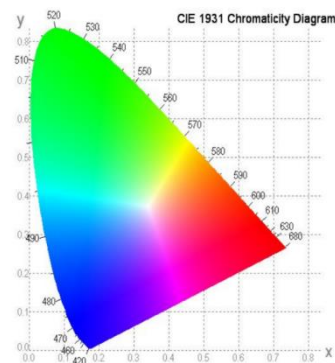
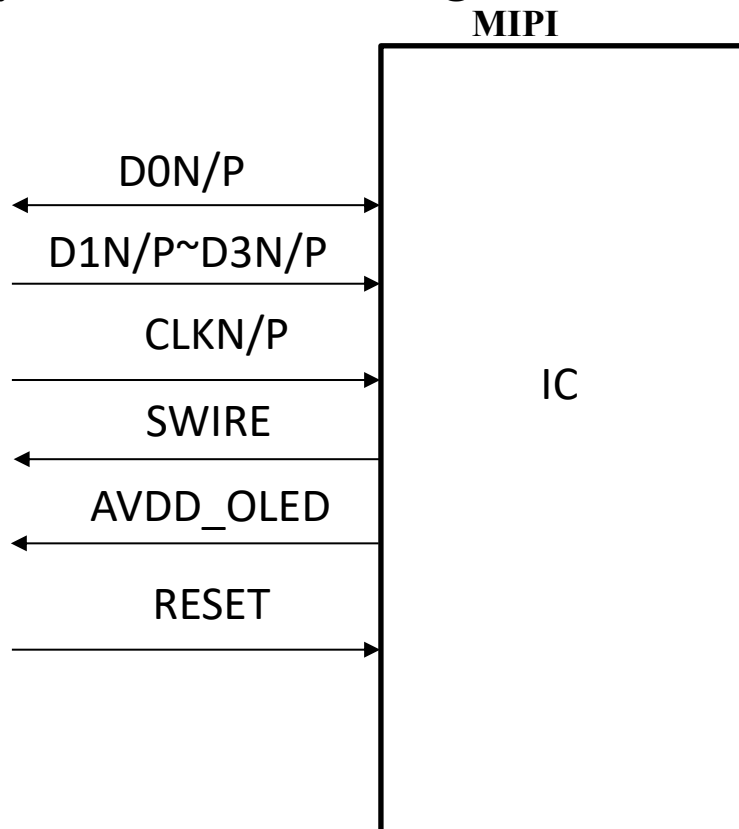


FIG.4. NTSC ratio 色域定义

7. Displayer Module Interface Description(Pin Definition)

NO.	SYMBOL	Description	I/O
1	VPP	Power supply for OTP. Leave the pin to open when not in use.	P
2	D3N	MIPI DSI data3-	I
3	NC	Not connected	-
4	D3P	MIPI DSI data3+	I
5	ELON2	Swire protocol setting pin	O
6	GND	The power ground	P
7	VDDP_EN	Power IC enable control pin	I
8	D0N	MIPI DSI data0-	I/O
9	PCD	Panel crack detection	O
10	D0P	MIPI DSI data0+	I/O
11	TE	Tearing effect output pin to synchronize MCU to frame writing, activated by S/W command.	O
12	GND	The power ground	P
13	ERR_FG	Error flag. Leave the pin to open when not in use.	P
14	CKN	MIPI DSI clock-	I
15	RESX	External Reset,Low is Active	I
16	CKP	MIPI DSI clock+	I
17	VDDIO	I/O Power Supply	P
18	GND	The power ground	P
19	TSP_1.8V	Not connected	-
20	D1N	MIPI DSI data1-	I
21	VIN_6.5V	AMOLED charge pumping	P
22	D1P	MIPI DSI data1+	I
23	VCI	AMOLED logic	P
24	GND	The power ground	P
25	TSP_SDA	Not connected	-
26	D2N	MIPI DSI data2-	I
27	TSP_SCL	Not connected	-
28	D2P	MIPI DSI data2+	I
29	TSP_ATTN	Not connected	-
30	GND	The power ground	P
31	TSP_3.3V	Not connected	-
32	TSP_RESET	Not connected	-
33	TSP_1.8V	Not connected	-
34	NC	Not connected	-
35	ELVDD	Power supply for Analog system	P
36	ELVSS	Power supply for Analog system	P
37	ELVDD	Power supply for Analog system	P
38	ELVSS	Power supply for Analog system	P
39	ELVDD	Power supply for Analog system	P
40	ELVSS	Power supply for Analog system	P

8. Displayer Module Block Diagram



9. Caution

9.1 Handling of Displayer Module

Be sure to ground the body when handling the displayer module.

Do not give external shock.

Do not apply excessive force on the surface.

Liquid in displayer is hazardous substance. Must not lick and swallow.

When the liquid is attach to your hand, skin, cloth etc. Wash it out thoroughly and immediately.

Do not operate it above the absolute maximum rating.

Do not disassemble the displayer module.

9.2 Storage

Store in an ambient temperature of 5°C to 45°C, and in a relative humidity of 40% to 60%. Do not expose to sunlight or intensive ultraviolet rays.

Storage in a clean environment, free from dust, active gas, and solvent.

Store in anti-static electricity container.

Store without any physical load.

10. Displayer Module Quality Criteria

10.1 Inspection Conditions

Inspection performed under the following conditions:

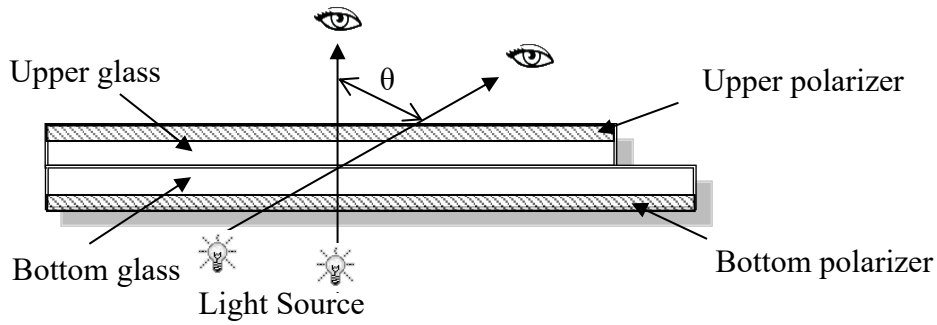
Temperature: $25 \pm 5^{\circ}\text{C}$

Humidity: $65\% \pm 10\%\text{RH}$

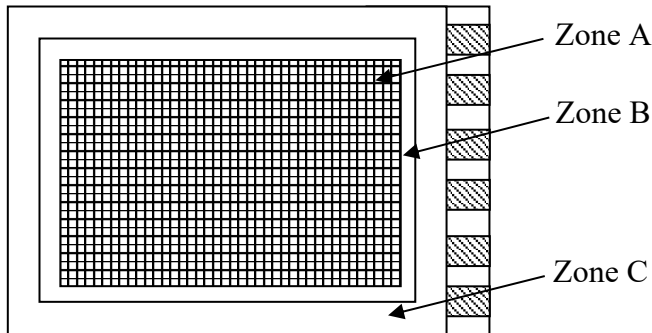
Viewing angle θ : $\leq 30^{\circ}$

Illumination: single fluorescent lamp (300 to 700Lux)

Viewing distance: $30\text{cm} \leq L \leq 50\text{cm}$



10.2 Zone Definition



Zone A : Effective viewing area(AA)

Zone B : Viewing area except Zone A

Zone C : Outside (Zone A+Zone B) which can not be seen after assembly by customer .)

Note: Visual defect in Zone C can be ignored, as it does not effect displayer function or appearance after being assembled by customer.

10.3 Inspection Specification

Sampling plan according to GB/T2828.1-2012/ISO 2859-1: 1999 and ANSI/ASQC

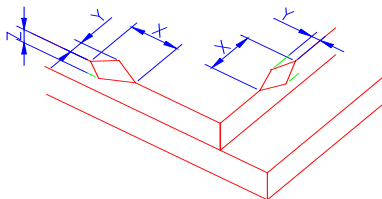
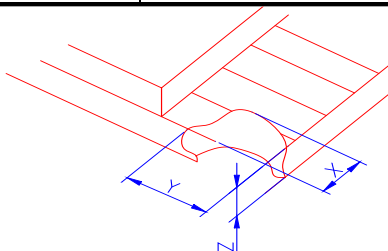
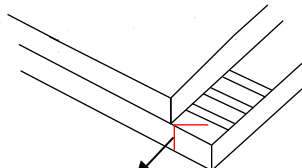
Z1.4-1993,normal level 2 and based on:

AQL:

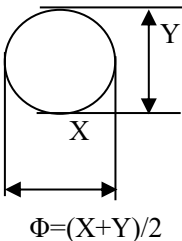
Major Defect	Minor Defect
0.4	1.0

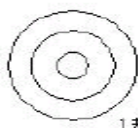
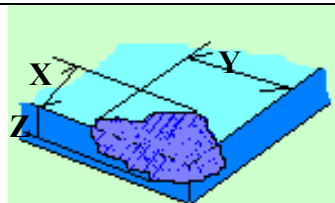
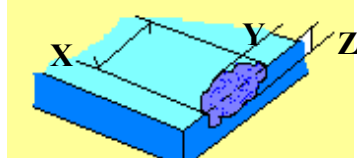
No.	Items to be inspected	Criteria	Classification of Defect
1	Functional Defects	1) No displayer, open or miss line. 2) Displayer abnormally, short. 3) Backlight no lighting, abnormal lighting. 4) TP no function.	Major
2	Missing	Missing component.	
3	Outline Dimension	Overall outline dimensions are larger or shorten than the drawing.	
4	Color Tone	Color unevenness, refer to limited sample.	Minor
5	Soldering Appearance	Good soldering , Peeling off is not allowed.	
6	Displayer/Polarizer/TP	Black/White spot/line, scratch, crack, etc.	
7	Linearity	No more than 1.5%.	

10.4 Criteria (Visual)

No.	Items	Criteria(mm)						
1	Displayer Edge Broken	 <table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td><Inner border line of the seal</td><td>≤T</td></tr></table> No effective function. Allowed	X	Y	Z	≤3.0mm	<Inner border line of the seal	≤T
	X	Y	Z					
	≤3.0mm	<Inner border line of the seal	≤T					
Displayer Corner Broken	 <table><tr><th>X</th><th>Y</th><th>Z</th></tr><tr><td>≤3.0mm</td><td>≤L</td><td>≤T</td></tr></table> No effective function. Allowed	X	Y	Z	≤3.0mm	≤L	≤T	
X	Y	Z						
≤3.0mm	≤L	≤T						
Displayer Crack	 Crack Not allowed							

Notes: X=Length of defect, Y=Width of defect, Z=Height of defect. L=Length of ITO, T=Height of displayer.

2	Clear Spot Defect  Φ=(X+Y)/2	1.light dot(Displayer/TP/Polarizer black/white spot , light dot, pinhole, dent, stain)																									
		<table><tr><th rowspan="2">Zone Size (mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.1</td><td colspan="3">Ignore</td></tr><tr><td>0.1<Φ≤0.2</td><td colspan="3">2(distance≥5mm)</td></tr><tr><td>Φ>0.2</td><td colspan="3">0</td></tr></table>	Zone Size (mm)	Acceptable Qty			A	B	C	Φ≤0.1	Ignore			0.1<Φ≤0.2	2(distance≥5mm)			Φ>0.2	0								
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		Φ≤0.1	Ignore																								
		0.1<Φ≤0.2	2(distance≥5mm)																								
		Φ>0.2	0																								
		2.Dim spot(Displayer/TP/Polarizer dim dot, light leakage、 dark spot)																									
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		Zone Size (mm)		Acceptable Qty																							
A	B		C																								
Φ≤0.1	Ignore																										
0.1<Φ≤0.2	2(distance≥5mm)																										
0.2<Φ≤0.25	1																										
Φ>0.25	0																										
3.TP Dirt																											
<table><tr><th rowspan="2">Zone Size (mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.1</td><td colspan="3">Ignore</td></tr><tr><td>0.1<Φ≤0.25</td><td colspan="3">2(distance≥5mm)</td></tr><tr><td>Φ>0.25</td><td colspan="3">0</td></tr></table>	Zone Size (mm)	Acceptable Qty			A	B	C	Φ≤0.1	Ignore			0.1<Φ≤0.25	2(distance≥5mm)			Φ>0.25	0										
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Φ>0.25	0																										
Line Defect	Black or white line, foreign material, scratch, stain on Displayer/TP/polarizer.																										
	<table><tr><th rowspan="2">Width(mm)</th><th rowspan="2">Length(mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.02</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr><tr><td>0.02<W≤0.03</td><td>L≤2.0</td><td colspan="2">N≤1</td></tr><tr><td>0.03<W≤0.05</td><td>L≤1.0</td><td colspan="2">N≤1</td></tr><tr><td>W>0.05</td><td colspan="4">Define as spot defect</td></tr></table>	Width(mm)	Length(mm)	Acceptable Qty			A	B	C	Φ≤0.02	Ignore	Ignore		Ignore	0.02<W≤0.03	L≤2.0	N≤1		0.03<W≤0.05	L≤1.0	N≤1		W>0.05	Define as spot defect			
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		A	B		C																						
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W>0.05	Define as spot defect																										
3	Polarizer Bubble	<table><tr><th rowspan="2">Zone Size (mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.1</td><td colspan="3">Ignore</td></tr><tr><td>0.1<Φ≤0.25</td><td colspan="3">2(distance≥5mm)</td></tr><tr><td>Φ>0.25</td><td colspan="3">0</td></tr></table>	Zone Size (mm)	Acceptable Qty			A	B	C	Φ≤0.1	Ignore			0.1<Φ≤0.25	2(distance≥5mm)			Φ>0.25	0								
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0.1<Φ≤0.25	2(distance≥5mm)																										
Φ>0.25	0																										
4	SMT	According to IPC-A-610C class II standard.Function defects and missing parts are major defects, others are minor defects.																									

5	Newton Ring	Newton Ring area>1/6 TP area NG Newton Ring area≤1/6 TP area OK			 1 规律性																						
		Rule-less Newton Ring area>1/4 TP area NG Rule-less Newton Ring area≤1/4 TP area OK			 似牛顿环  2 非规律性																						
	TP Corner Broken	<table><tr><td>X</td><td>Y</td><td>Z</td></tr><tr><td>X≤2.0mm</td><td>Y≤2.0mm</td><td>Z<Lens thickness</td></tr></table> * Circuitry broken is not allowed.	X	Y	Z	X≤2.0mm	Y≤2.0mm	Z<Lens thickness																			
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6	TP Dirt	<table><tr><th rowspan="2">Zone Size (mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.1</td><td colspan="3">Ignore</td></tr><tr><td>0.1<Φ≤0.2</td><td colspan="3">2(distance≥5mm)</td></tr><tr><td>Φ>0.2</td><td colspan="3">0</td></tr></table>				Zone Size (mm)	Acceptable Qty			A	B	C	Φ≤0.1	Ignore			0.1<Φ≤0.2	2(distance≥5mm)			Φ>0.2	0					
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0.1<Φ≤0.2	2(distance≥5mm)																										
Φ>0.2	0																										
7	Foreign Material on TP Film	The line can be seen after mobile phone in the operating condition <table><tr><th rowspan="2">Width(mm)</th><th rowspan="2">Length(mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.03</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="2">Ignore</td></tr><tr><td>0.03<W≤0.05</td><td>L≤3.0</td><td colspan="2">N≤2</td></tr><tr><td>W>0.05</td><td colspan="4">L>3 Define as spot defect</td></tr></table>				Width(mm)	Length(mm)	Acceptable Qty			A	B	C	Φ≤0.03	Ignore	Ignore		Ignore	0.03<W≤0.05	L≤3.0	N≤2		W>0.05	L>3 Define as spot defect			
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8	Dim Line on TP Film	<table><tr><th rowspan="2">Width(mm)</th><th rowspan="2">Length(mm)</th><th colspan="3">Acceptable Qty</th></tr><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>Φ≤0.03</td><td>Ignore</td><td colspan="2">Ignore</td><td rowspan="3">Ignore</td></tr><tr><td>0.03<W≤0.05</td><td>L≤3.0</td><td colspan="2">N≤2</td></tr><tr><td>W>0.05</td><td colspan="2">Define as spot defect</td></tr></table>				Width(mm)	Length(mm)	Acceptable Qty			A	B	C	Φ≤0.03	Ignore	Ignore		Ignore	0.03<W≤0.05	L≤3.0	N≤2		W>0.05	Define as spot defect			
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0.03<W≤0.05	L≤3.0	N≤2																									
W>0.05	Define as spot defect																										
9	Pattern Font	Pattern fonts are clear and symmetrical,slightly uneven are allowed.Font swing is no more than 1mm,font line should be smooth.																									

11. Reliability Test

No.	Item	Condition	Standard
1	High Temp. Storage	80℃, 48 hours	1. Functional test is OK. Missing Segment, short, unclear segment, displayer abnormally and liquid crystal leak are not allowed. 2. No low temperature bubbles, end seal loose and fall, frame rainbow.
2	Low Temp. Storage	-30℃, 48 hours	
3	High Temp. Operation	70℃, 48 hours	
4	Low Temp. Operation	-20℃, 48 hours	
5	High Temp. and High Humidity Storage	60℃, 90%RH, 48 hours	
6	Thermal and Cold Shock	Static state, -40℃(30 Min)~80℃(30 Min)~ -40℃(30Min), packaging, 10 cycles	1. Function test is OK. 2. No glass crack, chipped glass, end seal loose and fall, epoxy frame crack and so on. 3. No structure loose and fall.
7	Vibration Test	Packaging, Frequency : 10-55Hz Amplitude : 1.0mm, Each direction on X,Y,Z for 2 hours	
8	Dropping Test	Pack products into a carton box, and drop it to ground from 80cm height, once each side of the carton.	

Notes:

- 11.1 The reliability items will be fully carried out in new project sample-test.
11.2 The reliability items will be tested as a continuous monitoring during mass production by plan.
11.3 All samples are inspected after being kept in the room with normal temperature and humidity for 2 hours or above after reliability test.
11.4 Vibration test: It is not necessary without assembly frame of backlight, or PCB/FPC etc..
11.5 Dropping test: It is not necessary until confirm new package.
11.6 For the high temperature and high humidity test, pure water of over 10 MΩ.cm should be used.
11.7 Each test item applies only once for every test displayer. Tested displayer cannot be used again in any other test item.
11.8 The quantity of displayer for each reliability item is 5pcs to 10pcs.

12 Precautions for Using Displayer Modules

12.1 Safety Instructions

- 12.1.1 If the displayer panel breaks, be careful not to get any liquid crystal substance in your mouth.
12.1.2 If the liquid crystal substance touches your skin or clothes, please wash it off immediately by using soap and water.

12.2 Handling Precautions

- 12.2.1 Avoid static electricity damaging the displayer.
12.2.2 Do not remove the panel or frame from the module .
12.2.3 The polarizer of the displayer is very fragile, so please handle it very carefully.
12.2.4 Do not wipe the polarizer with a dry cloth, as it may easily scratch the surface of it.
12.2.5 The color tone of displayer has the possibility of being changed in the large storage temperature range.
12.2.6 Pay attention to the working environment, as the components may be destroyed by static electricity.
--Be sure to ground human body and electric appliance during working.
--Avoid working in a dry environment to minimize the generation of static electricity.
--Static electricity may be generated when the protective film is quickly peeled off.
12.2.7 When soldering the terminal of displayer, make sure the power source of soldering iron does not leak.
12.2.8 If the displayer surface contaminated, clean it gently with a clean cloth. If it is heavily contaminated, moisten the clean cloth with the following solvent (ex:ethyl alcohol). Solvents without mentioned may damage the polarizer.

12.3 Operation Instructions

12.3.1 It is recommended to drive the displayer within the specified voltage limits, try to adjust the operating voltage for the optimal contrast, the color and contrast of displayer panel will varies at different temperature.

12.3.2 Response time is greatly delayed at low operating temperature range. However, it does not mean the displayer is damaged, it will recover in the specified temperature range.

12.3.3 If the displayer area is pushed hard during operation, the displayer will become abnormal.

12.3.4 Do not operate the displayer at the environments over the specified conditions, this may cause damage on the displayer and shorten the lifetime.

12.4 Storage Instructions:

12.4.1 Store displayer in a sealed polyethylene bag.

12.4.2 Store displayer in a dark place, Do not expose to sunlight or fluorescent light. Keep the temperature between 0°C and 35°C.

12.4.3 Avoid the polarizer being touched by any objects. (Recommend to store in the container while shipped)

12.5 Limited Warranty

12.5.1 AML will replace or repair any of its displayer module, after customers inspected defects according to AML's quality standard (in this specification), with a period of 12 months from ink-print date on product.

12.5.2 Any defect product must be returned to AML within 60 days since shipped out. Confirmation of such date shall be based on freight documents. The warranty liability of AML limited to repair and/or replacement on defects above (10.4)

12.5.3 No warranty can be granted if the precautions stated above have been disregarded. The typical samples are as below:

--Displayer glass cracked or broke.

--Displayer modified by grinding or engraving or painting varnished etc..

--PCB/FPC components damaged or modified.

--PCB/FPC cracked or damaged.

--Protect film of polarizer removed.

12.5.4 Modules must be returned with sufficient descriptions of defects. Any connectors or cable installed by the customer must be removed completely without damaging the displayer. Modules must be packed with the container in which they were shipped. The figures below illustrate the correct and incorrect packaging.



13. Packing Specification

-TBD-